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Fiber-reinforced origami electronics with high rigidity and flexibility for display applications



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Origami structures provide functional advantages to rigid electronics through geometric transformations. However, the transformations involved in folding and deployment cause stress concentration on flexure hinges of origami structure, triggering electronic malfunction. Here, we report origami electronics based on a fiber-reinforced electronic composite. A thin PEDOT:PSS-based electronic composite minimizes stress during folding without electrode damage. Nylon is embedded in this foldable composite and, despite being thin and flexible for folding, provides high tensile resistance to prevent plastic deformation and tearing under tension. This strategy enables the creation of flexure hinges for origami electronics that maintain mechanical and electrical stability under repeated transformations. Origami electronics that integrate the high-durability composite can be used in display applications supporting 25-fold compression with the Flasher origami structure and 2D-to-3D deployment with the Kresling origami structure. The ability of origami electronics to withstand bending and tensile stress enables shape-reconfigurable displays requiring repeated reconfiguration across multiple hinges.

Origami structures, which are capable of forming various three-dimensional (3D) structures via simple folding of a two-dimensional (2D) plane, offer an approach for diversifying the forms of electromechanical systems^{1–5}. These 3D structures are determined by the design and arrangement of flexure hinges or crease-type hinges, which are flexible under external forces^{6,7}. This strategy allows the realization of physical expandability and a more pronounced volumetric effect, in contrast to the limitations of 2D structures. For example, depending on the pattern, the dense arrangement of hinges enables compression and expansion by several hundred times for the Flasher pattern or the construction of a 3D shape for the Kresling pattern^{8–11}. When integrated with electronic components, such origami structures can have potential applications in various fields, including biomedical applications requiring spatial efficiency, soft robots engineered for locomotion without complex structures, and electronic devices designed to transform into 3D configurations^{12–18}.

In origami-based electromechanical systems, the hinge is thin to minimize the bending stress applied to the embedded electrode during folding, thus preventing electronic malfunction and mechanical damage. Flexible electronics with low stiffness can serve as the hinge in origami electronics through adaptation of the bending stress^{19,20}. These electronics

are typically composed of a conductive layer, such as a metal film or nanowires, deposited on a soft or thin substrate, which ensures high durability against folding^{20–24}. Nevertheless, such flexibility and softness of the electrode make it vulnerable to tensile forces, which can stretch or tear it. This vulnerability, caused by low stiffness, may limit the stability of the hinge in origami electronics during repeated tension in deployment.

The limited durability of the hinge in such origami electronics highlights the need for hinge reinforcement. Recent research into origami electronics has revealed the potential to achieve enhanced durability of the fragile hinge. For example, flexible wood-based origami electronics provide high tensile strength and foldability, laying the foundation for embedding electronics into origami structures²⁵. Nevertheless, the relatively low durability under repeated folding may limit the stable performance of origami electronics, which are subjected to at least several thousand folding cycles. An approach based on the stiffness modulation of hinges via the phase transition of smart materials (e.g., shape memory polymers) can simultaneously ensure durability under both tension and folding^{26,27}. However, the external stimulus (e.g., Joule heating) required to induce hinge flexibility consumes a significant amount of energy, limiting the feasibility of long-term operation.

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Fiber reinforcement is a method that preserves the inherent outstanding properties of a material while compensating for its relatively weak characteristics^{28,29}. This approach can serve as an effective strategy for ensuring durability against both folding and tension, which is an essential requirement for origami-based electronics. For example, nylon, which is a type of fiber, possesses high tensile resistance due to its robust and resilient characteristics, which enable it to resist strong pulling forces and, upon elongation, absorb excess stress.

Here, we present origami electronics based on a fiber-reinforced electronic flexure hinge that can endure folding and pulling during repeated deployment. To achieve the conflicting material properties, the flexure hinge consists of an electronic composite based on polydimethylsiloxane (PDMS) and poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) with tough nylon embedded. Before integrating this flexure hinge into an origami structure, we optimized the thickness and spacing of the rigid links of the origami to be connected by the flexure hinge, which reduced the stress on the flexure hinge during unfolding. The origami electronics integrated with the flexure hinge maintained their conductivity after repeated in-folding and out-folding cycles because PEDOT:PSS coated on flexible PDMS. Additionally, the embedded nylon absorbed the tensile forces generated during deployment, preventing the origami electronics from stretching, as confirmed by finite element analysis (FEA) simulations and cyclic tensile tests. This origami electronic structure, highly durable against external forces, was integrated with a touch panel and a light-emitting diode (LED), demonstrating its potential for display applications. Furthermore, application of this strategy to dense origami patterns with angular designs (e.g., Flasher and Kresling patterns) demonstrated that the origami electronics reliably operate without mechanical or electrical malfunction during complex geometric transformations. This structurally simple fiber-reinforcement strategy, which simultaneously achieves the conflicting mechanical properties of foldability and tensile strength, provides a foundation for ensuring mechanical and electrical stability during shape transformations of origami electronics, where deformation occurs at numerous origami hinges.

Results

Overview of origami electronics based on a fiber-reinforced flexure hinge

The schematic illustration in Fig. 1a shows the origami electronics integrating a fiber-reinforced electronic composite designed to withstand bending and tensile stress during folding and deployment. The polymer electrode layer in the composite is composed of PEDOT:PSS, which consists of hole-transporting PEDOT and dispersed network-forming PSS³⁰. PEDOT chains form linear interconnected pathways anchored to the PSS matrix, enabling stable electrical conduction under repeated bending deformation^{31,32}. Additionally, the nylon fibres embedded in the composite as fibre reinforcement prevent irreversible stretching and tearing while maintaining the current flow during deployment, accompanied by pulling. This highly durable composite is integrated with rigid links to form the origami electronics, enabling folding and deployment according to the designed origami pattern (Fig. 1b). The exposed electronic composite between the rigid links acts as a flexure hinge within the origami electronics. The folding stress in the folding origami electronics can become concentrated on the flexure hinge, with mechanical and electrical damage. However, the flexure hinge of PEDOT:PSS, sandwiched by a thin encapsulation layer (polyurethane) and a substrate (PDMS), allows the origami electronics to maintain LED operation without electrical failure, thus exhibiting a higher foldability than that of other flexible electrodes, including Ag nanowires and thin Au films (Fig. 1c). This thin flexure hinge may seem vulnerable to pulling during deployment of the origami electronics, but the relatively high toughness of nylon can prevent tearing of the hinge by absorbing strain energy (Fig. 1d). Such origami electronics with a high resistance to folding and pulling allow various applications that utilize mechanical shape transformation. For example, as shown in Fig. 1e, the dense arrangement and angular design of the hinges enable the electronics

to be expandable and portable. High compression of Flasher pattern-based origami electronics can expose numerous hinges to stress, but does not induce resistance changes associated with mechanical or electrical damage. Another hinge pattern design for the origami electronics involves a structure that transforms between 2D and 3D configurations (Fig. 1f). Kresling pattern-based origami electronics can be utilized for wearable electronics, offering storage efficiency in an easily foldable state and enabling touch-based machine-human interactions with users when deployed. During such repeated shape transformation and tactile input, the origami electronics do not exhibit electronic malfunction. The detailed fabrication process of the origami electronics is described in the Supplementary Fig. 1. The detailed calculation and analysis of strain energy absorption are described in the Supplementary Note 1 and Supplementary Fig. 2.

Optimization of the folding configuration for origami electronics

Figure 2a shows the layer structure of the fabricated origami electronics based on smart composite microstructures that enable folding through rigid links based on polyethylene terephthalate (PET) and the fiber-reinforced flexure hinges^{6,33}. As shown in the scanning electron microscopy (SEM) image in Fig. 2b, the PET layer (~200 µm) is attached to the fiber-reinforced electronic composite based on PEDOT:PSS (~400 nm) encapsulated in polyurethane (~20 µm) and PDMS (~150 µm) using an optically clear adhesive (OCA) (~50 µm) to form origami electronics. The relatively high thickness and stiffness (Young's modulus: 2.8–3.1 GPa)³⁴ of the PET layer can prevent bending during folding of the origami electronics (Supplementary Fig. 3). Additionally, the elastic PEDOT:PSS embedded in the electronic composite, which serves as the electronic component of the origami electronics, offers mechanical durability against repeated folding. Here, laser etching of PET exposes the electronic composite to form a flexure hinge for folding of the origami electronics. This folding consists of mountain folding (Out-folding) and valley folding (In-folding) (Fig. 2c). Compared with valley folding, mountain folding toward a rigid link can induce greater strain because of potential collisions between rigid links. Such collisions may lead to electrode disconnection accompanied by stretching of the flexure hinge. Figure 2d shows the optimal gap distance between rigid links to prevent excessive strain on the flexure hinge during mountain folding. Narrow gaps (200, 400, and 600 µm) lead to electrode disconnection during folding, whereas a relatively wide gap (800 µm) prevents excessive strain. The folding stability with the optimized gap is further demonstrated by cycling tests under mountain folding. As shown in Fig. 2e, the origami electronics maintain stable electric resistance over repeated mountain folding (~20,000 cycles) without collisions between rigid links. Additionally, valley folding of the origami electronics does not damage the electrode because of the inherent flexibility of the flexure hinge, despite a relatively narrow gap (~200 µm) (Fig. 2f).

In addition to the gap between rigid links, stress concentration occurs at a vertex where multiple flexure hinges meet. This stress concentration becomes even more significant as the pattern of the origami electronics increases in complexity because of the convergence of more flexure hinges at a single vertex. Figure 2g shows the Miura and Waterbomb origami patterns, which incorporate both mountain folding and valley folding motions and a vertex with four and six hinges, respectively. To minimize stress concentration at the vertex, we performed additional laser etching at that location. The optimized gap and removed vertex, as shown in Fig. 2h and Supplementary Movie 1, allow the origami electronics to maintain stable electrical resistance after the initial stabilization phase (~50 cycles). The resistance increase observed during stabilization originates from the plastic deformation of the polyurethane layer and electrode damage occurring at the interface, where the electrode is replaced by nylon, including under repeated folding. However, such damage is confined to a limited area and does not disrupt the current path required for the operation of the electronics (Supplementary Fig. 4). Additionally, the results of repeated folding cycle tests (10,000 cycles) of the Miura and Waterbomb origami patterns shown in Fig. 2i, Supplementary Figs. 5 and 6 demonstrate that gap optimization and vertex removal are effective approaches for integrating

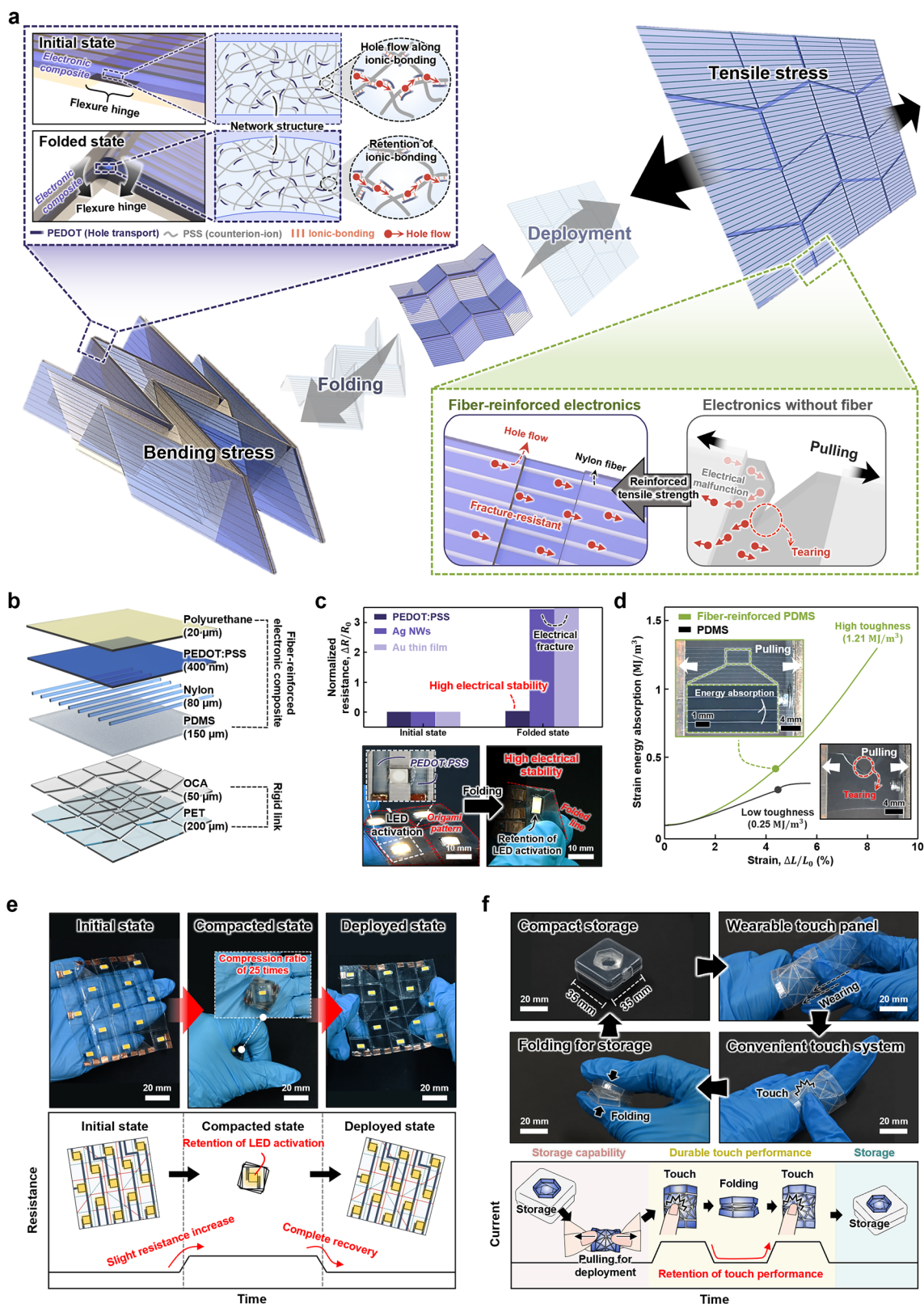


Fig. 1 | Origami electronics based on a fiber-reinforced electronic composite. **a** Schematic illustration of origami electronics with high-durability flexure hinges for repeated folding and deployment. **b** Exploded-view schematic illustration of the origami electronics. **c** Normalized resistance ($\Delta R/R_0$) of PEDOT:PSS, Ag NWs, and an Au film in the half-folded state, and corresponding images of a PEDOT:PSS-based LED circuit demonstrating electrically stable operation. **d** Strain energy

absorption of fiber-reinforced PDMS and PDMS under tension. **e** Photograph of origami electronics applied as a 25-fold compressible display, and a schematic illustration of the electrical stability during folding and deployment. **f** Photograph of a storage-capable ring-shaped 3D touch panel application, and a schematic illustration of the electrical stability during operation.

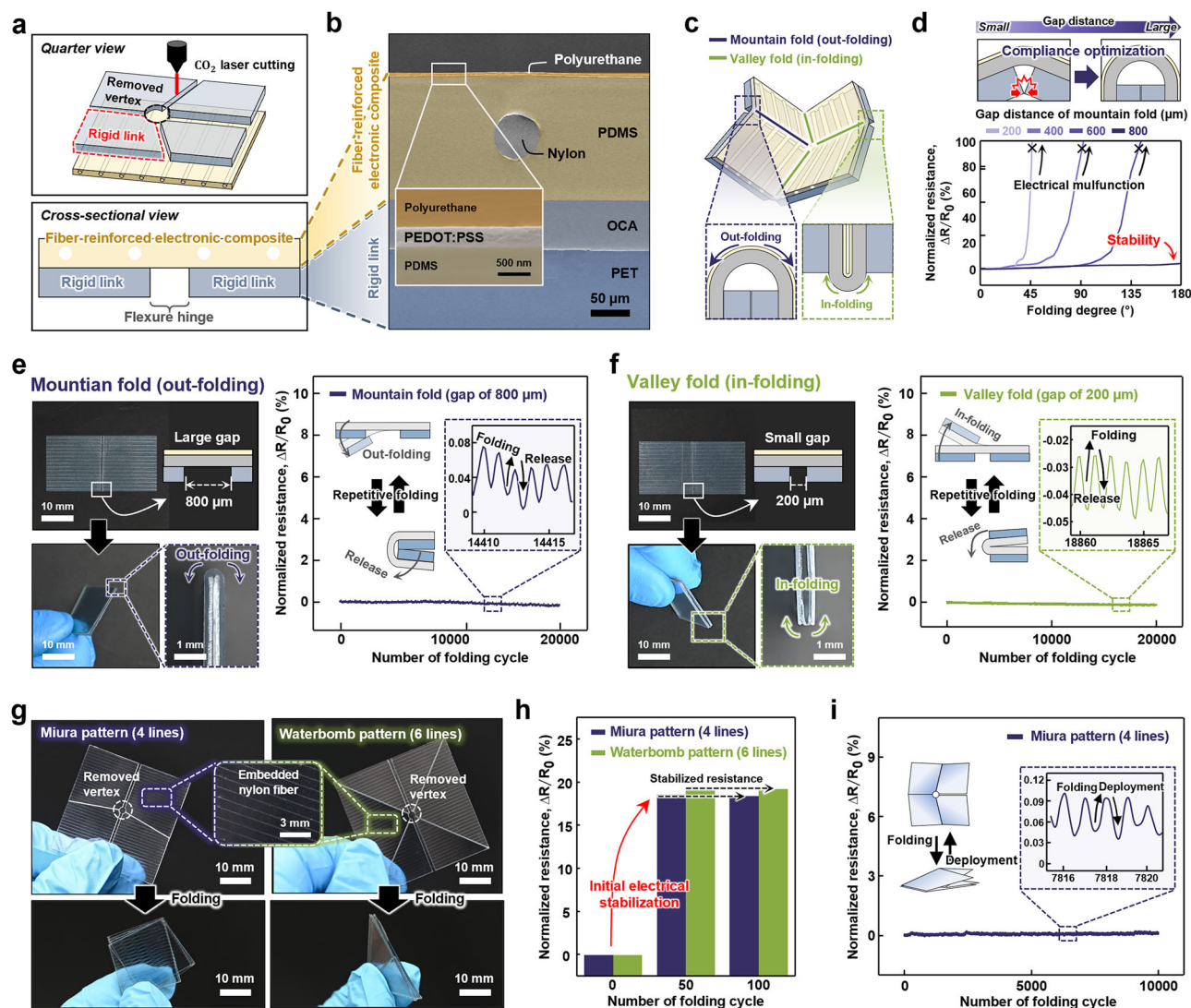


Fig. 2 | Mechanical and electrical characterization of fiber-reinforced origami electronics during folding. **a** Schematic illustration of the layered structure of the origami electronics. **b** Cross-sectional SEM image of the origami electronics. **c** Schematic illustration of two folding types in the origami electronics: mountain folding and valley folding. **d** Comparison of the electrical stabilities (normalized resistance, $\Delta R/R_0$) during mountain folding with various flexure hinge gap distances (200 μm , 400 μm , 600 μm , and 800 μm). **e** Photographs of mountain folding, and

measured resistance over 20,000 mountain folding cycles. **f** Photographs of valley folding, and measured resistance over 20,000 valley folding cycles. **g** Photographs of origami electronics with the Miura pattern (4 flexure hinges) and the Waterbomb pattern (6 flexure hinges), along with an image showing vertex removal. **h** Comparison of resistances before folding and after 50 and 100 folding cycles for the Miura (purple) and Waterbomb (green) patterns. **i** Measured resistances of origami electronics with the Miura pattern during 10,000 folding cycles.

embedded electronics into repeatedly folded origami structures and are not limited to specific patterns.

Tensile optimization and analysis of origami electronics

A flexure hinge with low rigidity and a small thickness in origami electronics is effective in minimizing folding-induced stress, but its inherent flexibility and softness may lead to stretching or tearing. As a result, it becomes vulnerable to strong pulling forces applied by the user during deployment. The tough nylon embedded in the fabricated origami electronics can prevent irreversible deformation induced by tensile forces (Supplementary Movie 2). Additionally, dense embedding of the nylon enhances the resistance to tensile forces. The schematic image in Fig. 3a illustrates the calculation process for determining the nylon embedding ratio within the electronic composite (fiber volume fraction, V_f) on the basis of its density. As shown in Fig. 3b and Supplementary Fig. 7, increasing the amount of nylon fiber in the flexure hinge (V_f : 0.75%, 1%, 1.5%, and 3%) allows the origami electronics to withstand a fracture force proportional to the fiber content (7.9 N, 11.4 N, 12.5 N, and 22.1 N), indicating enhanced tensile

resistance. Additionally, such durability during pulling can guarantee electrical reliability and prevent mechanical fracture. Figure 3c presents the measured resistance of origami electronics corresponding to the V_f conditions shown above under tensile loading. When a tensile force is applied to the origami electronics, a relatively high V_f (3%) enables the structure to maintain electrical resistance under large tensile loads (up to ~ 20 N).

To confirm the performance of the optimized origami electronics (V_f : 3%) for high resistance to pulling, as shown in the schematic sequence in Fig. 3d, pulling was simulated via FEA. Nylon can absorb tensile stress during pulling of the origami electronics (Supplementary Fig. 8a). This stress absorption prevents excessive stretching of the flexure hinge with low stiffness by minimizing the stress (~ 22.2 MPa) transmitted to the hinge. In contrast, the origami electronics without embedded nylon exhibit a high stress concentration (~ 48.2 MPa) at the flexure hinge (Fig. 3e). Additionally, Nylon can also absorb shear and torsional forces induced by sliding and twisting, enhancing the durability of the origami electronics under complex deformation (Supplementary Fig. 8b, c). Fiber reinforcement can prevent failure due to complex deformations, such as sliding and twisting, that may

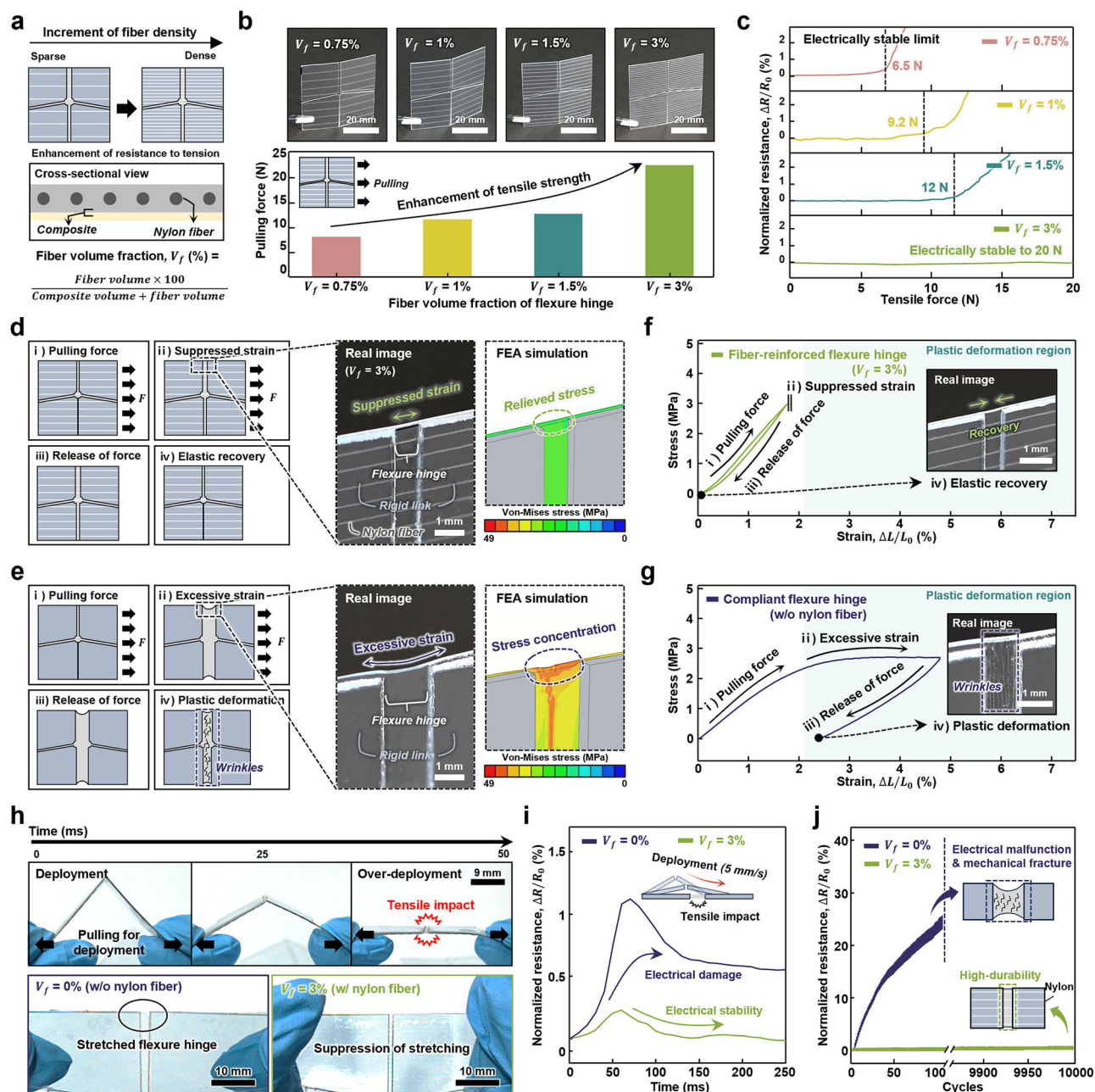


Fig. 3 | Mechanical and electrical characterization of fiber-reinforced origami electronics during deployment. **a** Schematic illustration of the fiber volume fraction (V_f , %) and corresponding formula for V_f in the fiber-reinforced origami electronics. **b** Fracture force under uniaxial tension of the origami electronics depending on the fiber volume fraction (V_f : 0.75%, 1%, 1.5% and 3%). **c** Comparison of the resistance changes in the origami electronics with various fiber volume fractions after uniaxial tension. **d** Schematic illustration of the elastic behavior of flexure hinges in the origami electronics under applied tension, accompanied by a photograph and an FEA simulation of the flexure hinge. **e** Schematic illustration of the plastic deformation process in the flexure hinges of the fiber-free origami electronics

under excessive strain, accompanied by a photograph and an FEA simulation. **f** Results of a uniaxial tensile test on the origami electronics with nylon reinforcement. **g** Results of a uniaxial tensile test on the origami electronics without nylon reinforcement. **h** Time sequence of the tensile impact test on the flexure hinge (50 ms), with magnified images of deformation without and with nylon reinforcement ($V_f = 0\%$ and 3%). **i** Measured resistance of the origami electronics without and with nylon reinforcement during tensile impact tests (speed = 5 mm/s). **j** Measured resistance of the origami electronics with and without nylon reinforcement under repeated tensile impact tests (~10,000 cycles).

occur during origami folding and deployment (Supplementary Fig. 9). Such origami electronics demonstrating stress relief in the simulation achieve limitation of the strain to within the elastic deformation range of the hinge, as shown in the strain-stress curve in Fig. 3f. However, application of the same tensile stress to the origami electronics without nylon leads to plastic deformation at the flexure hinge (Fig. 3g). The origami electronics can be exposed not only to such tensile forces during deployment but also to tensile impacts as the deployment speed increases, causing damage due to over-

deployment accompanied by overstretching. For example, as shown in Fig. 3h, when the user rapidly deploys the origami electronics (~50 ms), the flexure hinge can undergo irreversible stretching, but the embedded nylon prevents this deformation. Additionally, the embedded electrodes in the origami electronics can be vulnerable to such tensile impacts. As shown in Fig. 3i and Supplementary Fig. 10, rapid dynamic deployment (applied at ~5 mm/s) of the origami electronics induces an increase in the electrical resistance (~0.6%), in contrast to the structure with nylon reinforcement.

Such a slight increase in the electrical resistance may appear negligible, but continuous exposure to such tensile impacts can lead to progressive damage to the electrode, resulting in complete failure of the electronic functionality of the origami electronics. Figure 3j shows the results of repeated tensile impact tests (~10,000 cycles) on the origami electronics, comparing structures with and without nylon reinforcement. The origami electronics with nylon embedded maintain electrical stability by absorbing the tensile impacts during repeated deployment, whereas the accumulated tensile impacts on the origami electronics without nylon reinforcement cause electrical failure, accompanied by a change in resistance (~23.2%).

Incorporation of display components into origami structures

The fiber-reinforced origami electronics can be utilized for display applications that incorporate both a touch panel and LED components. When the Miura origami pattern is applied to the structure, it enables the realization of a highly space-efficient display (compression ratio: 8.2 times) (Fig. 4a and Supplementary Fig. 11)³⁵. The fabricated Miura origami-based electronics integrate a touch panel, which is one of the components of the display system. The touch panel used operates via a surface capacitive mechanism^{24,36}, forming a uniform electric field across the embedded PEDOT:PSS electrode layer. When an object contacts the surface, the origami electronics perceive the touch by detecting a disruption in the electric field. For example, when a user touches the surface of the origami-based touch panel, the disturbance in the electric field induces a current flow from the edge electrodes to the touch point, which enables the exact location of the touch to be determined (Supplementary Fig. 13a). A touch adjacent to the electrode (A1) connected to the current meter, as shown in Fig. 4b, induces a relatively high current change (touch point (TP)#1: ~37.3%), whereas a touch farther from A1 results in a lower current change (TP#2: ~5.1%, TP#3: ~2.8%, TP#4: ~4.8%). This distance-dependent current variation enables localization of the touch point (Supplementary Fig. 13b–d). Such distance-dependent identification can remain reliable during folding (30% and 90%) of the origami electronics, which enables stable touch input under dynamic deformation conditions (Fig. 4c and Supplementary Fig. 14). When this folding and deployment deformation is repeated, it may disturb the drawing input on the touch panel because of fatigue degradation of the hinge (Fig. 4d). However, the origami electronics designed to withstand folding and pulling continue to detect writing on the touch panel after multiple folding (~100 times) and deployment (100 times) cycles (Fig. 4e, f). While these folding and deployments generally involve external forces applied in predetermined directions according to the origami pattern, practical handling can subject the electronics to multidirectional forces. For example, the origami electronics maintain electrical durability under pulling deformation in vertical or diagonal direction even, twisting deformation (Supplementary Fig. 15). Further details of the performance evaluation of the touch panel based on the PEDOT:PSS electrode are provided in Supplementary Note 2.

We also integrated LEDs into the origami electronics to provide light emission, which is a fundamental function of a display. Figure 4g shows that the PEDOT:PSS-based patterned circuit passes through the flexure hinge. This patterned electrode, as shown in Fig. 4h, allows the emission of multiple LEDs (~16 units) to be modulated, enabling the generation of diverse visual patterns. Although a relatively narrow circuit pattern (~3 mm) for selectively controlling each LED can lead to disconnection during folding and pulling, the origami electronics form diverse patterns (letters: M, O, S, T) through controlled LEDs (Fig. 4i). The cycling test in Fig. 4j emphasizes the electronic durability of the origami electronics under repeated deformation. When multiple folding and pulling cycles (~100 times) are applied, the origami electronics continue to receive a stable current (~1.65 mA) without circuit damage (Fig. 4k). Such maintenance of the electronic functionality under narrow circuit layouts in the origami structure provides the foundation for reconfiguration of electronics into various shapes. Further details of the design of LED circuits based on PEDOT:PSS are provided in Supplementary Note 3 and Supplementary Fig. 16.

Highly compressible displays and 3D wearable electronics

The arrangement of flexure hinges defines the deployable configuration of the origami electronics. In this configuration, an increase in the number of hinges and the application of an angle design allow the origami electronics to perform more complex and precise mechanical functions. A high hinge density can cause continuous stress related to folding and pulling in regions with densely packed hinges. However, this challenge can be addressed by the proposed approach based on a fiber-reinforced flexure hinge. For example, Flasher pattern-based origami electronics can offer both portability and expandability (~25 times) to the user through a dense pattern and angle control of the flexure hinge (Fig. 5a and Supplementary Fig. 17)^{37,38}. Multiple LEDs were embedded in Flasher origami electronics to enable display functionality (Supplementary Movie 3). Figure 5b shows the compression process of the Flasher origami electronics, while the letter “O” is displayed to verify the stability of the electronic functionality during folding. Upon compression around the central core origami cell, the bending stress may appear to concentrate at the dense hinges, but the LEDs continue to display the letter without a change in the applied current. Such compressed origami electronics experience tensile stress in deployment, but their toughness allows them to maintain output without current changes (Fig. 5c). The detailed optimization of the high compression ratio of the Flasher pattern-based origami structure is provided in Supplementary Note 4.

Another example of a complex pattern of flexure hinges presents a potential application in wearable electronics using 3D shape transformation of the Kresling pattern. A densely repeated Kresling pattern (cylindrical structure) composed of pairs of triangular facets enables the transformation of electronics from a 2D to 3D structure (Fig. 5d)^{39,40}. This pattern modulates the folding resistance, allowing the cylindrical structure to either be compressed under gentle pressure or maintain its 3D shape under a relatively strong pressing force. For example, the cylindrical structure formed by a triangular pattern with a 25° angle exhibits highly compliant structural behavior (Supplementary Fig. 18a). In contrast, a relatively large angle of 65° leads to a high-rigidity structure that resists folding under a pressing force (~500 g) (Supplementary Fig. 18b). These two angles (25° and 65°), when used in triangular patterns as interior angles, provide a bistable structural strategy that allows a cylindrical structure to be selectively tuned for easy foldability and structural robustness. Figure 5e presents the sequential folding and deployment process (100 cycles) of the origami electronics designed with the Kresling patterns with defined angles (25° and 65°). The initial resistance of the embedded electrode is preserved throughout the repetitive deformation process. This electrical stability is also effective in the high-rigidity mode, in which the origami electronics withstand heavy loads, preserving their electrical functionality and shape (Fig. 5f). Such Kresling-patterned origami electronics can be utilized as wearable electronics worn on a finger, enabling remote control of a smartphone (Fig. 5g). The high compressibility of the wearable electronics allows for portability and storability before they are worn. Once deployed, the wearable electronics can adapt to a finger and stably maintain their structural configuration under the forces applied during touch. As shown in Fig. 5h, sequential deployment and compression of the wearable electronics do not affect their touch-sensing performance. This stability of the electronic function enables the wearable electronics to be used to respond to an incoming call through a sliding touch motion (Fig. 5i and Supplementary Movie 4). A detailed analysis of each state of the bistable 3D cylindrical structure is provided in Supplementary Note 5.

Discussion

In this study, we developed fiber-reinforced origami electronics capable of geometric transformation without fatigue degradation. The main strategy to prevent this degradation relies on a durable flexure hinge integrating a flexible electronic composite and tough nylon. These different mechanical advantages minimize bending and tensile stresses on the flexure hinge, respectively, during the repeated shape transformation of origami electronics. Such stability against repeated stresses realizes the potential for deployable displays through the availability of numerous flexure hinge arrangements and angular configurations. The large-range

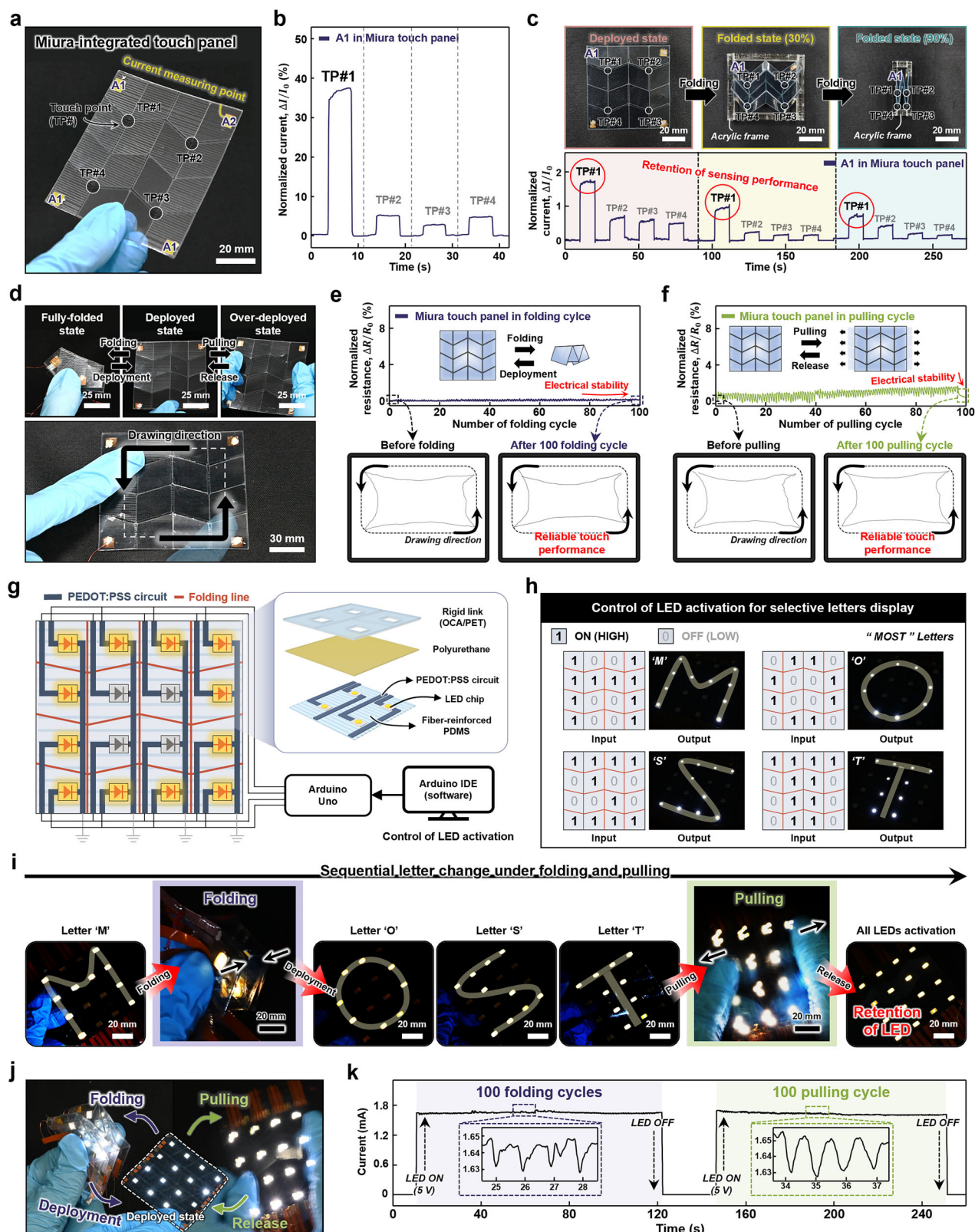


Fig. 4 | Integration of display components into fiber-reinforced origami electronics. **a** Photograph of the origami touch panel based on the Miura pattern. **b** Current responses at TP#1–TP#4 detected from A1 upon touch input on the touch panel. **c** Comparison of the currents at TP#1–TP#4 detected from A1 upon touch input in the deployed (red), 30% folded (yellow), and 90% folded (green) states. **d** Photographs of the folded and deployed touch panel. **e** Reliable drawing input on the touch panel without electrical failure after 100 folding cycles. **f** Reliable drawing input on the touch panel without electrical failure after 100 pulling cycles.

g Schematic illustration of LED integration based on Miura pattern-based origami electronics. **h** Schematic illustration of the selective control of an LED array (4 × 4) for letter display in the origami electronics. **i** Pattern formation (letters: M, O, S, T, all) through sequential control during folding and pulling processes. **j** Photographs of the folding/pulling process of an LED array based on the origami electronics. (All 16 LEDs are on.) **k** LED current measured during 100 repeated folding and pulling cycles.

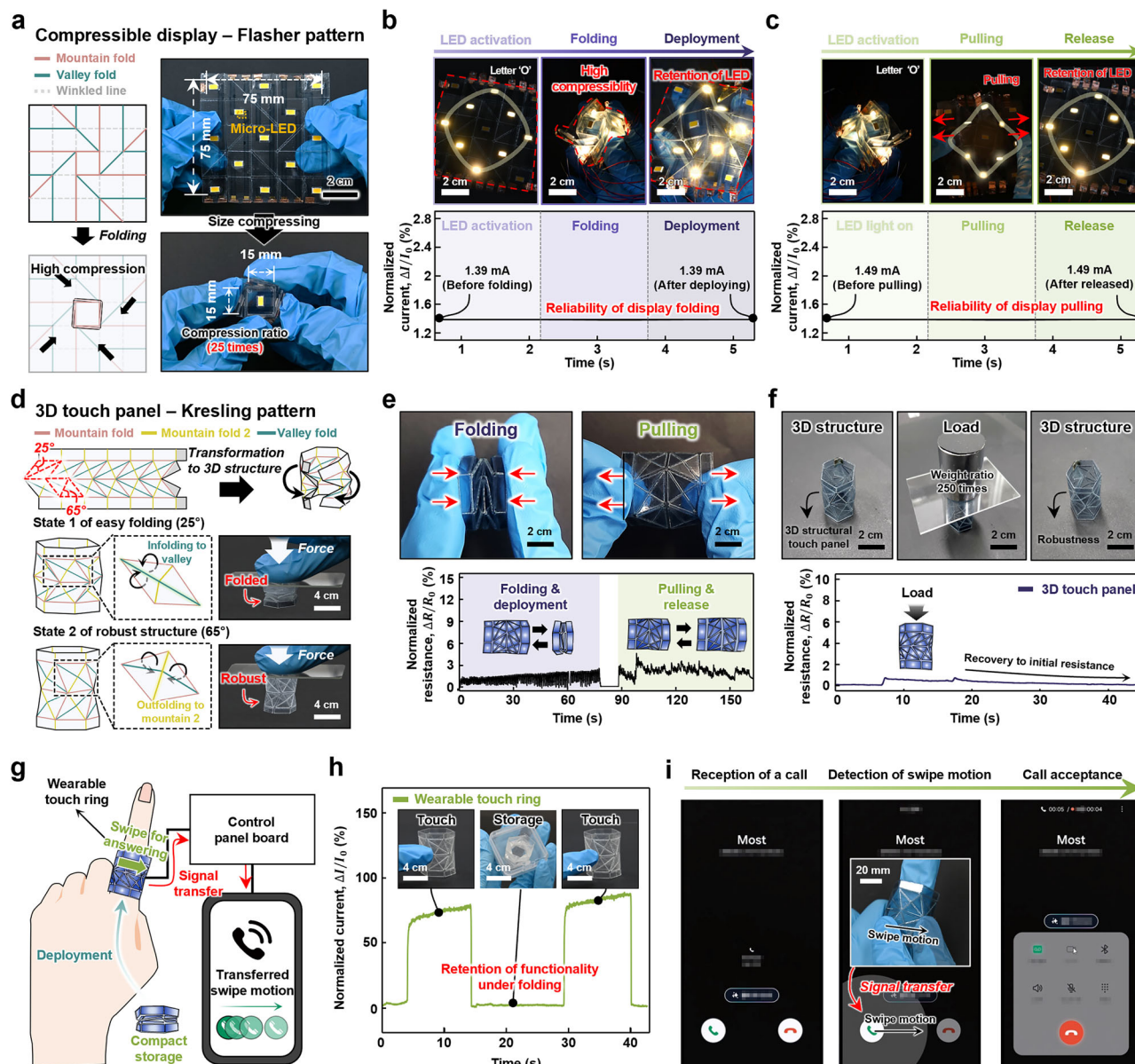


Fig. 5 | Application of the fiber-reinforced origami electronics in deployable display systems. a LED array based on fiber-reinforced origami electronics with a Flasher pattern enabling 25-fold compression. **b** Photographs and measured current ($\Delta I/I_0$) during folding of the Flasher pattern-based origami electronics. **c** Photographs and measured current during pulling of the Flasher pattern-based origami electronics. **d** Bistable Kresling-patterned 3D touch panel based on the

origami electronics. **e** Measured resistance of the 3D touch panel during repeated folding and pulling cycles. **f** Measured resistance of the 3D touch panel under an external force (250 times the weight) in the robust state (state 2). **g** Schematic illustration of the 3D touch panel used as a wearable touch ring. **h** Measured current of the 3D touch panel in the deployed and storage states. **i** Sequential images of the user interaction with the 3D touch panel for call acceptance.

compression–expansion and 2D-to-3D transformation of the origami electronics demonstrate the potential of the origami electronics for various display applications based on shape reconfiguration. The current application for this strategy focuses on shape-reconfigurable displays, but when an actuator (e.g., a thin polymer-based actuator)^{41–43} is integrated into the flexure hinge layer, its use could be extended to robotic applications that incorporate both actuation and sensing functionalities.

Methods

Procedure for fabricating the origami electronics

The nylon fibers were fitted into the uniformly spaced grooves of an acrylic plate (85 × 90 mm), followed by oxygen plasma treatment (100 W, 30 s). Polydimethylsiloxane (PDMS) solution (Sylgard™ 184; K1 Solution, Gwangmyeong, Republic of Korea) was spin-coated onto the nylon-aligned acrylic plate and cured in an oven at 80 °C for 3 h. The poly (3,4-

ethylenedioxythiophene): poly(styrenesulfonate) (PEDOT:PSS) solution (655201; Sigma-Aldrich, St. Louis, MO, USA) was spin-coated onto the fibre-embedded PDMS and cured at 70 °C for 3 h. The composite was treated with oxygen plasma (100 W, 30 s), and a thin metal layer consisting of 2 nm Cr and 90 nm Au was deposited via sputtering. The polyurethane resin for the protective layer was spin-coated over the electrode layer and cured at 90 °C for 12 h. For the fabrication of the origami electronics, a 50 µm OCA was attached to 200 µm PET film, and the laminated film was laser-cut along the designed origami pattern. Finally, the OCA surface of the rigid links was bonded to the fibre-reinforced electronic composite.

FEA of the neutral plane and strain on the electrode when bending the electronic composite

3D FEA simulations for determining the tensile stress distribution in the flexure hinges were constructed using ABAQUS software. The elements of

the flexure hinges, composed of rigid links and the electronic composite, were implemented as C3D8R (8-node linear brick, reduced integration, hourglass control). Both the fibre-reinforced and non-reinforced flexure hinges were expected to be linear elastic materials. The Young's modulus and Poisson's ratio of these flexure hinges were as follows: $E_{\text{nylon}} = 5.075$ GPa, $\nu_{\text{nylon}} = 0.3$, $E_{\text{rigid_link}} = 4.5$ GPa, $\nu_{\text{rigid_link}} = 0.33$, $E_{\text{PDMS}} = 1.4$ MPa, $\nu_{\text{PDMS}} = 0.48$, $E_{\text{polyurethane}} = 1.2$ GPa, and $\nu_{\text{polyurethane}} = 0.33$.

Folding cycle tests of the origami electronics

Repeated folding experiments of the origami electronics were conducted using a linear actuator (LSM6-NK235630 Linear Motor, Motorbank, South Korea). Acrylic folding supports were mounted on the two ends of the actuator to serve as fixtures for the folding process. The test specimen was secured between the supports using adhesive tape, and repeated folding motions were induced by driving the linear actuator. The displacement and speed of the actuator were precisely controlled using a control board (Arduino Mega 2560, Arduino, Italy) in combination with a DC-DC buck converter module (LM2596, generic, China) and Arduino IDE software. A total of 10,000 to 20,000 folding cycles was performed. During the experiment, the electrical characteristics of the specimen, such as the changes in its resistance, were monitored in real time using a data acquisition (DAQ) system (DEWEsoft).

Pulling cycle test of the origami electronics in deployment

Tensile tests were conducted to investigate electrode failure caused by tensile stress on the flexure hinge during the unfolding of origami electronics. First, to observe the stretching behavior of the flexure hinge, the two ends of the origami electronics were held, and the structure was unfolded at a speed of 200 mm/s. The unfolding process was recorded using a high-speed digital camera (Phantom MIRO EX4 and MIRO C320, Vision Research, USA) operating at 1000 frames per second to accurately capture the unfolding time. To evaluate the electrical performance of the origami electronics under rapid and repeated unfolding conditions, cyclic tensile tests were performed using a universal testing machine (3342 UTM, Instron Co., Norwood, MA, USA) at a speed of 5 mm/s for 100 cycles. During the tests, the electrical signal between the two electrodes of the origami electronics was continuously monitored in real time using a DAQ system (DEWEsoft).

Data availability

All the data generated or analyzed during this study are included in this published article and its supplementary information files, or from the corresponding author upon reasonable request.

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Author contributions

D.G., M.K., and S.H. contributed equally to this work. D.G. performed the design and engineering investigation of the origami structure. M.K. contributed to the design and experiments of origami electronics with integrated micro-LED circuits. S.H. contributed to the design and experiments of 3D touch panels based on origami structures. J.J. and I.H.

manufactured the electronic composite. Y.R. and S.H. supervised the project. All authors reviewed and approved the final manuscript.

Competing interests

All the authors declare that they have no financial or nonfinancial competing interests.

Additional information

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